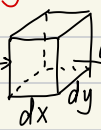
# Continuity Equation

$\rho u dydz \rightarrow$  $\rightarrow (\rho u + \frac{\partial(\rho u)}{\partial x} dx) dydz$



$\dot{m}_{net,x} = \dot{m}_{in,x} - \dot{m}_{out,x} = -\frac{\partial(\rho u)}{\partial x} dxdydz = -\frac{\partial(\rho u)}{\partial x} d\forall$

Similarly: $\dot{m}_{net,y} = -\frac{\partial(\rho v)}{\partial y} d\forall$ ; $\dot{m}_{net,z} = -\frac{\partial(\rho w)}{\partial z} d\forall$

$\dot{m}_{net} = \dot{m}_{net,x} + \dot{m}_{net,y} + \dot{m}_{net,z} = -(\frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} + \frac{\partial(\rho w)}{\partial z}) d\forall$

$m = \rho d\forall \quad \frac{\partial m}{\partial t} = \frac{\partial \rho}{\partial t} d\forall$

According to the conservation of mass:

Mass Accumulation = Net Mass Inflow

$\therefore \frac{\partial \rho}{\partial t} d\forall = -(\frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} + \frac{\partial(\rho w)}{\partial z}) d\forall$

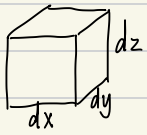
Finally:

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} + \frac{\partial(\rho w)}{\partial z} = 0$$

# Navier-Stokes Equation

Newton's Second Law:

$\vec{F} = ma \quad m = \rho \forall \quad \forall = dxdydz$





$\vec{F} = \rho dxdydz \vec{a}$

$\vec{a} = \frac{d\vec{v}}{dt}$ $\vec{v}$ is the function of $x, y, z, t$

According to the chain rule: $\vec{a} = \frac{d\vec{v}(x,y,z,t)}{dt} = \frac{\partial \vec{v}}{\partial t} + \frac{\partial v}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial v}{\partial y}\frac{\partial y}{\partial t} + \frac{\partial v}{\partial z}\frac{\partial z}{\partial t}$

$u = \frac{\partial x}{\partial t}, \ v = \frac{\partial y}{\partial t} \quad w = \frac{\partial z}{\partial t} \quad \therefore \vec{a} = \frac{\partial \vec{v}}{\partial t} + u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} + w\frac{\partial v}{\partial z} = \frac{\partial \vec{v}}{\partial t} + (\vec{v} \cdot \vec{\nabla}) \cdot \vec{V} = \frac{D\vec{v}}{Dt}$

Pressure Force:

$F_{p,x} = -\frac{\partial p}{\partial x} d\forall \rightarrow \Sigma F_p = -(\frac{\partial p}{\partial x} + \frac{\partial p}{\partial y} + \frac{\partial p}{\partial z}) d\forall = -\nabla p d\forall$

Viscous Force:

$F_{u,x} = \mu \nabla^2 u \rightarrow \Sigma F_u = \mu(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 w}{\partial z^2})$

Body Force

$F_{g_x} = m\vec{g}_x = \rho d\forall g_x \rightarrow \Sigma F_g = \rho d\forall (g_x + g_y + g_z)$

Navier-Stokes Equation on X-axis:

$-\frac{\partial p}{\partial x} d\forall + u\frac{\partial^2 u}{\partial x^2} + \rho d\forall g_x = \rho d\forall (\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} + w\frac{\partial u}{\partial z})$

Rearrange: $\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} + w\frac{\partial u}{\partial z} = -\frac{1}{\rho}\frac{\partial p}{\partial x} + \nu\frac{\partial^2 u}{\partial x^2} + g_x$

# Pressure Poisson Equation

2D N-S Equation:

$$\begin{cases} \frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = -\frac{1}{\rho}\frac{\partial p}{\partial x} + \nu\nabla^2 u \\ \frac{\partial v}{\partial t} + u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} = -\frac{1}{\rho}\frac{\partial p}{\partial y} + \nu\nabla^2 v \end{cases}$$

Continuity Equation:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

Derive X:

$$\frac{\partial}{\partial x}\left(\frac{\partial u}{\partial t}\right) + \frac{\partial}{\partial x}\left(u\frac{\partial u}{\partial x}\right) + \frac{\partial}{\partial x}\left(v\frac{\partial u}{\partial y}\right) = -\frac{1}{\rho}\frac{\partial^2 p}{\partial x^2} + \frac{\partial}{\partial x}(\nu\nabla^2 u) \quad (1)$$

$$\frac{\partial}{\partial x}\left(u\frac{\partial u}{\partial x}\right) + \frac{\partial}{\partial x}\left(v\frac{\partial u}{\partial y}\right) : \left(\frac{\partial u}{\partial x}\right)^2 + u\frac{\partial^2 u}{\partial x^2} + \frac{\partial v}{\partial x}\frac{\partial u}{\partial y} + v\frac{\partial^2 u}{\partial x \partial y}$$

Derive Y:

$$\frac{\partial}{\partial y}\left(\frac{\partial v}{\partial t}\right) + \frac{\partial}{\partial y}\left(u\frac{\partial v}{\partial x}\right) + \frac{\partial}{\partial y}\left(v\frac{\partial v}{\partial y}\right) = -\frac{1}{\rho}\frac{\partial^2 p}{\partial y^2} + \frac{\partial}{\partial y}(\nu\nabla^2 v) \quad (2)$$

$$\frac{\partial}{\partial y}\left(u\frac{\partial v}{\partial x}\right) + \frac{\partial}{\partial y}\left(v\frac{\partial v}{\partial y}\right) : \frac{\partial u}{\partial y}\frac{\partial v}{\partial x} + u\frac{\partial^2 v}{\partial x \partial y} + \left(\frac{\partial v}{\partial y}\right)^2 + v\frac{\partial^2 v}{\partial y^2}$$

(1) + (2):

LHS: $\frac{\partial}{\partial x}\left(\frac{\partial u}{\partial t}\right) + \frac{\partial}{\partial y}\left(\frac{\partial v}{\partial t}\right) + \left(\frac{\partial u}{\partial x}\right)^2 + u\frac{\partial^2 u}{\partial x^2} + \frac{\partial v}{\partial x}\frac{\partial u}{\partial y} + v\frac{\partial^2 u}{\partial x \partial y} + \frac{\partial u}{\partial y}\frac{\partial v}{\partial x} + u\frac{\partial^2 v}{\partial x \partial y} + \left(\frac{\partial v}{\partial y}\right)^2 + v\frac{\partial^2 v}{\partial y^2}$

$\frac{\partial}{\partial x}\left(\frac{\partial u}{\partial t}\right) + \frac{\partial}{\partial y}\left(\frac{\partial v}{\partial t}\right) = \frac{\partial}{\partial t}\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}\right)$ According to the Continuity Equation, this term equals 0.

The purple term: $(u_x)^2 + u \cdot u_{xx} + v_x \cdot u_y + v \cdot u_{xy} + u_y \cdot v_x + u \cdot v_{xy} + (v_y)^2 + v \cdot v_{yy}$

Continuity Equation: $u_x + v_y = 0$

Derive Continuity Equation with X: $u_{xx} + v_{yx} = 0$

Derive Continuity Equation with Y: $u_{xy} + v_{yy} = 0$

So the purple term $= (u_x)^2 + 2u_y v_x + (v_y)^2$

RHS: $-\frac{1}{\rho}\frac{\partial^2 p}{\partial x^2} - \frac{1}{\rho}\frac{\partial^2 p}{\partial y^2} + \frac{\partial}{\partial x}(\nu\nabla^2 u) + \frac{\partial}{\partial y}(\nu\nabla^2 v)$

$\frac{\partial}{\partial x}(\nu\nabla^2 u) + \frac{\partial}{\partial y}(\nu\nabla^2 v) = \nu\nabla^2\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}\right)$ According to the Continuity Equation, this term equals 0.

Final Pressure Poisson Equation:

$$\frac{\partial^2 p}{\partial x^2} + \frac{\partial^2 p}{\partial y^2} = -\rho\left[\left(\frac{\partial u}{\partial x}\right)^2 + 2\frac{\partial u}{\partial y}\frac{\partial v}{\partial x} + \left(\frac{\partial v}{\partial y}\right)^2\right]$$

$$\therefore \nabla^2 p = b \qquad b = -\rho\left[\frac{\partial^2 u}{\partial x^2} + 2\frac{\partial u}{\partial y}\frac{\partial v}{\partial x} + \frac{\partial^2 v}{\partial y^2}\right]$$

In the CFD codes:

$$b = \rho\left[\frac{1}{\Delta t}\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}\right) - \frac{\partial^2 u}{\partial x^2} - 2\frac{\partial u}{\partial y}\frac{\partial v}{\partial x} - \frac{\partial^2 v}{\partial y^2}\right]$$

# Finite-Difference Formulation of the Pressure Poisson Source Term b:

$$\frac{\partial u}{\partial x} = \frac{u_{i,j+1} - u_{i,j-1}}{2\Delta x} \qquad \frac{\partial u}{\partial y} = \frac{u_{i+1,j} - u_{i-1,j}}{2\Delta y}$$

$$\frac{\partial v}{\partial x} = \frac{v_{i,j+1} - v_{i,j-1}}{2\Delta x} \qquad \frac{\partial v}{\partial y} = \frac{v_{i+1,j} - v_{i-1,j}}{2\Delta y}$$

$$b_{i,j} = \rho\left[\frac{1}{\Delta t}\left(\frac{u_{i,j+1} - u_{i,j-1}}{2\Delta x} + \frac{v_{i+1,j} - v_{i-1,j}}{2\Delta y}\right) - \left(\frac{u_{i,j+1} - u_{i,j-1}}{2\Delta x}\right)^2 - 2\left(\frac{u_{i+1,j} - u_{i-1,j}}{2\Delta y}\right)\left(\frac{v_{i,j+1} - v_{i,j-1}}{2\Delta x}\right) - \left(\frac{v_{i+1,j} - v_{i-1,j}}{2\Delta y}\right)^2\right]$$